

1 Fundamental Inverse Trig Identities

We start with a simple, yet useful theorem.

Theorem 1. *The inverse of an odd function is odd.*

Proof. Suppose f is an odd function with inverse f^{-1} . Then for some $y = f(x)$,

$$\begin{aligned} f^{-1}(-y) &= f^{-1}(-f(x)) \\ &= f^{-1}(f(-x)) && \text{(as } f \text{ is odd)} \\ &= -x \\ &= -f^{-1}(y). \end{aligned}$$

Thus, f^{-1} is odd. □

There are several basic identities the Extension 1 student should be familiar with and can use/prove on demand. I list them below, and provide their proofs. It is also important to practise inverse trig **without** using the "right angled triangle" method, as it is unrigorous.

1. $\sin^{-1}(-x) = -\sin^{-1} x$
2. $\tan^{-1}(-x) = -\tan^{-1} x$
3. $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$
4. $\cos^{-1}(-x) + \cos^{-1} x = \pi$
5. $\tan^{-1} x + \tan^{-1} \frac{1}{x} = \begin{cases} \pi/2 & \text{if } x > 0 \\ -\pi/2 & \text{if } x < 0 \end{cases}$

Proof of 1. As $\sin^{-1}$ is the inverse of the odd function $\sin x$ restricted to the domain $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, by Theorem 1, $\sin^{-1}$ is odd and thus $\sin^{-1}(-x) = -\sin^{-1} x$. □

Proof of 2. Follows by Theorem 1. □

Proof of 3. Let $x = \cos y$ for some $y \in [0, \pi]$. By the complementary angle identity, we have

$$x = \sin\left(\frac{\pi}{2} - y\right),$$

and since $\frac{\pi}{2} - y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ which is the range of $\sin^{-1}$, applying $\sin^{-1}$ to both sides yields

$$\sin^{-1} x = \frac{\pi}{2} - y.$$

But $y = \cos^{-1} x$, completing the proof. □

Proof of 4. By the third identity, we have the following identities:

$$\bullet \cos^{-1}(-x) = \frac{\pi}{2} - \sin^{-1}(-x)$$

$$\bullet \cos^{-1} x = \frac{\pi}{2} - \sin^{-1} x$$

Adding the above together, we obtain

$$\begin{aligned} \cos^{-1}(-x) + \cos^{-1} x &= \pi - \sin^{-1}(-x) - \sin^{-1} x \\ &= \pi - (-\sin^{-1} x) - \sin^{-1} x && \text{(as } \sin^{-1} \text{ is odd)} \\ &= \pi, \end{aligned}$$

thus proving our claim. $\square$

Proof of 5. Define the function

$$f(x) = \tan^{-1} x + \tan^{-1} \frac{1}{x}, \quad (x \neq 0).$$

As f is differentiable for all $x \neq 0$ and its derivative is given by

$$f'(x) = \frac{1}{1+x^2} - \frac{1}{x^2} \cdot \frac{x^2}{x^2+1} = 0,$$

the function f must be piecewise-constant on all intervals it is continuous on, that is,

$$f(x) = \begin{cases} c_1 & \text{if } x > 0 \\ c_2 & \text{if } x < 0 \end{cases}$$

for some constants $c_1, c_2 \in \mathbb{R}$. By substituting values $x = 1$ and $x = -1$, we find that $c_1 = \frac{\pi}{2}$ and $c_2 = -\frac{\pi}{2}$, thus proving our claim. $\square$

2 Further Inverse Trig Identities

The Extension 1 student will find this handful of identities useful to recall and prove on demand. These should also be used in place of any of the right-angled triangle doctrine taught at schools. Once again, I list them below and provide their proofs. At the end of this section, I provide some examples of their application, and leave a few practise problems for the reader.

$$1. \cos(\sin^{-1} x) = \sqrt{1 - x^2}$$

$$2. \sin(\cos^{-1} x) = \sqrt{1 - x^2}$$

$$3. \sec(\tan^{-1} x) = \sqrt{1 + x^2}$$

$$4. \sin(\tan^{-1} x) = \frac{x}{\sqrt{1 + x^2}}$$

$$5. \tan^{-1} x + \tan^{-1} y = \tan^{-1} \left(\frac{x + y}{1 - xy} \right) \quad \text{for } xy < 1$$

$$6. \tan^{-1} x - \tan^{-1} y = \tan^{-1} \left(\frac{x - y}{1 + xy} \right) \quad \text{for } xy > -1$$

Proof of 1. By the Pythagorean Identity,

$$\begin{aligned} \cos^2(\sin^{-1} x) + \sin^2(\sin^{-1} x) &= 1 \\ \cos^2(\sin^{-1} x) &= 1 - x^2 \\ |\cos(\sin^{-1} x)| &= \sqrt{1 - x^2}. \end{aligned}$$

But $\sin^{-1} x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, and $\cos$ is non-negative for all values in this interval. Thus, we may drop the absolute value bars to obtain

$$\cos(\sin^{-1} x) = \sqrt{1 - x^2},$$

as required. □

Proof of 2. Identical to Proof of 1. □

Proof of 3. Left as an exercise. Hint: use the identity $\sec^2 \theta = 1 + \tan^2 \theta$. □

Proof of 4. We begin by writing the equation $\tan(\tan^{-1} x) = x$ as

$$\frac{\sin(\tan^{-1} x)}{\cos(\tan^{-1} x)} = x,$$

then upon dividing both sides by $\sec(\tan^{-1} x) = \sqrt{1 + x^2}$ (the third identity), we obtain

$$\sin(\tan^{-1} x) = \frac{x}{\sqrt{1 + x^2}}.$$

□

The following lemma will be useful to prove the fifth identity.

Lemma 1. *If $xy < 1$, then $-\frac{\pi}{2} < \tan^{-1} x + \tan^{-1} y < \frac{\pi}{2}$.*

Proof. Let $\alpha = \tan^{-1} x$ and $\beta = \tan^{-1} y$. Since $-\frac{\pi}{2} < \alpha, \beta < \frac{\pi}{2}$, we must have

$$-\pi < \alpha + \beta < \pi.$$

Our goal now is to tighten this inequality. By the cos compound angle identity,

$$\begin{aligned} \cos(\alpha + \beta) &= \cos \alpha \cos \beta - \sin \alpha \sin \beta \\ &= \frac{1 - xy}{\sqrt{1 + x^2} \sqrt{1 + y^2}} && \text{(by identities 3 and 4)} \\ &> 0. && \text{(as } xy < 1) \end{aligned}$$

The subset of $[-\pi, \pi]$ where cos is positive is $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, so it follows that

$$-\frac{\pi}{2} < \alpha + \beta < \frac{\pi}{2}.$$

□

Proof of 5. Let $\alpha = \tan^{-1} x$ and $\beta = \tan^{-1} y$, where $xy < 1$. By the tan compound angle identity,

$$\begin{aligned} \tan(\alpha + \beta) &= \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} \\ &= \frac{x + y}{1 - xy}. \end{aligned}$$

By Lemma 1, $\alpha + \beta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ which is the range of $\tan^{-1}$, so applying $\tan^{-1}$ to both sides yields

$$\tan^{-1} x + \tan^{-1} y = \tan^{-1} \left(\frac{x + y}{1 - xy} \right),$$

thus proving our claim. □

Proof of 6. This follows from the previous identity by replacing y with $-y$. □

Example 1. Prove that $\tan(\sin^{-1} x) = \frac{x}{\sqrt{1-x^2}}$.

Solution 1. Start by writing $\tan(\sin^{-1} x) = \frac{\sin(\sin^{-1} x)}{\cos(\sin^{-1} x)}$. Since $\sin(\sin^{-1} x) = x$ and as previously proven $\cos(\sin^{-1} x) = \sqrt{1-x^2}$, we swiftly arrive at the result

$$\tan(\sin^{-1} x) = \frac{x}{\sqrt{1-x^2}}.$$

Example 2. Solve the equation $\sin^{-1}(2x) = \tan^{-1}(5x)$.

Solution 2. As both sides of the equation are in the interval $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ and $\sin$ is one-to-one on this interval, we may apply $\sin$ to both sides to obtain the equivalent equation,

$$2x = \frac{5x}{\sqrt{1+25x^2}}.$$

Clearly $x = 0$ is a solution. In the case where $x \neq 0$, squaring both sides and simplifying yields the quadratic equation

$$100x^2 - 21 = 0,$$

and so our solutions are $x = 0, \frac{\sqrt{21}}{10}, -\frac{\sqrt{21}}{10}$.

Below are some practise problems for the kind reader! As per the spirit of this document, do not use the blasphemous right-angled triangle!

Problem 1.

(i) Prove that $\tan\left(\cos^{-1} \frac{1}{x}\right) = \frac{x\sqrt{x^2-1}}{|x|}$.

(ii) Hence, sketch the graph of $y = \tan\left(\cos^{-1} \frac{1}{1+x}\right)$.

Problem 2. Solve the equation

$$\tan^{-1}(2x) + \tan^{-1}(3x) = \frac{\pi}{4}.$$

Problem 3. Prove that $\sin^{-1} \sqrt{1-x^2} + |\sin^{-1} x| = \frac{\pi}{2}$.

Problem 4.

(i) Prove that for $x^2 + y^2 \leq 1$,

$$\sin^{-1} x + \sin^{-1} y = \sin^{-1} \left(x\sqrt{1-y^2} + y\sqrt{1-x^2} \right).$$

(ii) Hence, prove that for $-\frac{1}{2} \leq x \leq \frac{1}{2}$,

$$3 \sin^{-1} x = \sin^{-1}(3x - 4x^3).$$

3 The Out-of-Syllabus Functions and a Common Error

The $\cot^{-1} x$, $\sec^{-1} x$ and $\operatorname{cosec}^{-1} x$ functions are out-of-syllabus. I won't go into them in this document because they are rarely used, and it is not necessary to know them for the HSC. It is a matter of trivia to know that these functions can be defined as:

- $\cot^{-1} x = \frac{\pi}{2} - \tan^{-1} x$
- $\sec^{-1} x = \cos^{-1} \frac{1}{x}$
- $\operatorname{cosec}^{-1} x = \sin^{-1} \frac{1}{x}$

I often see students write $\cot^{-1} x = \tan^{-1} \frac{1}{x}$ for all non-zero real x . This is in fact false, due to the nuanced definition of $\cot^{-1}$. This statement holds if and only if $x > 0$.